EX 7 LINEAR AND LOGISTIC REGRESSION

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
df = pd.read_csv("/content/Housing.csv")
df.head()
df.info()
df.describe()
```

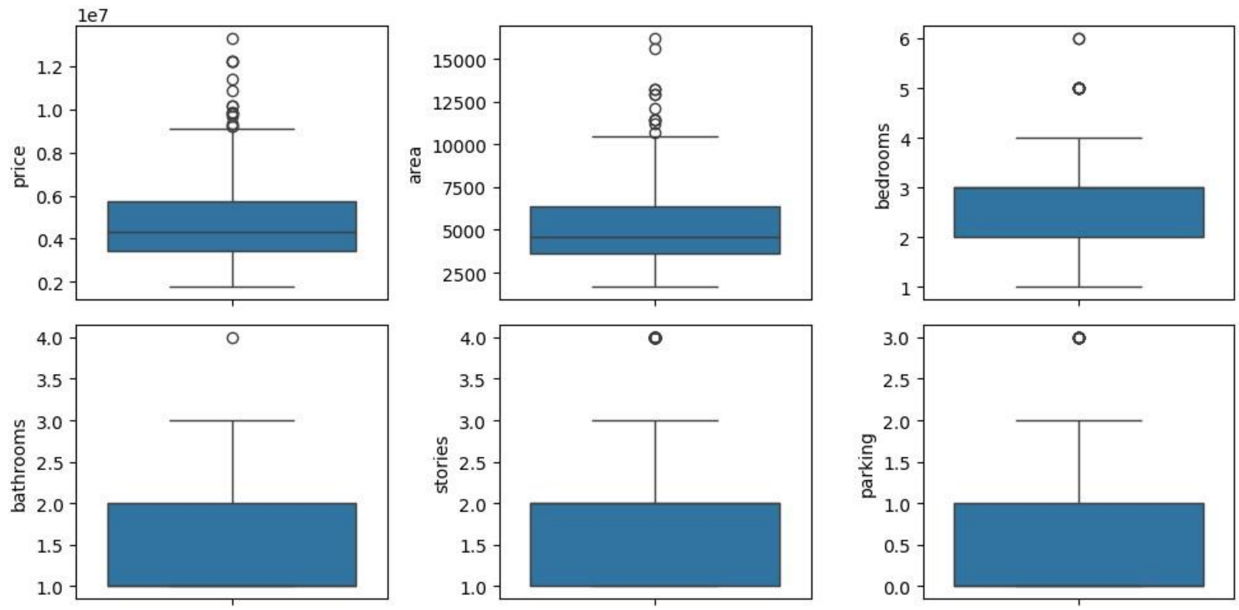
```
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   price             545 non-null    int64 
 1   area              545 non-null    int64 
 2   bedrooms          545 non-null    int64 
 3   bathrooms         545 non-null    int64 
 4   stories           545 non-null    int64 
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64 
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
```

	price	area	bedrooms	bathrooms	stories	parking
count	5.450000e+02	545.000000	545.000000	545.000000	545.000000	545.000000
mean	4.766729e+06	5150.541284	2.965138	1.286239	1.805505	0.693578
std	1.870440e+06	2170.141023	0.738064	0.502470	0.867492	0.861586
min	1.750000e+06	1650.000000	1.000000	1.000000	1.000000	0.000000
25%	3.430000e+06	3600.000000	2.000000	1.000000	1.000000	0.000000
50%	4.340000e+06	4600.000000	3.000000	1.000000	2.000000	0.000000
75%	5.740000e+06	6360.000000	3.000000	2.000000	2.000000	1.000000

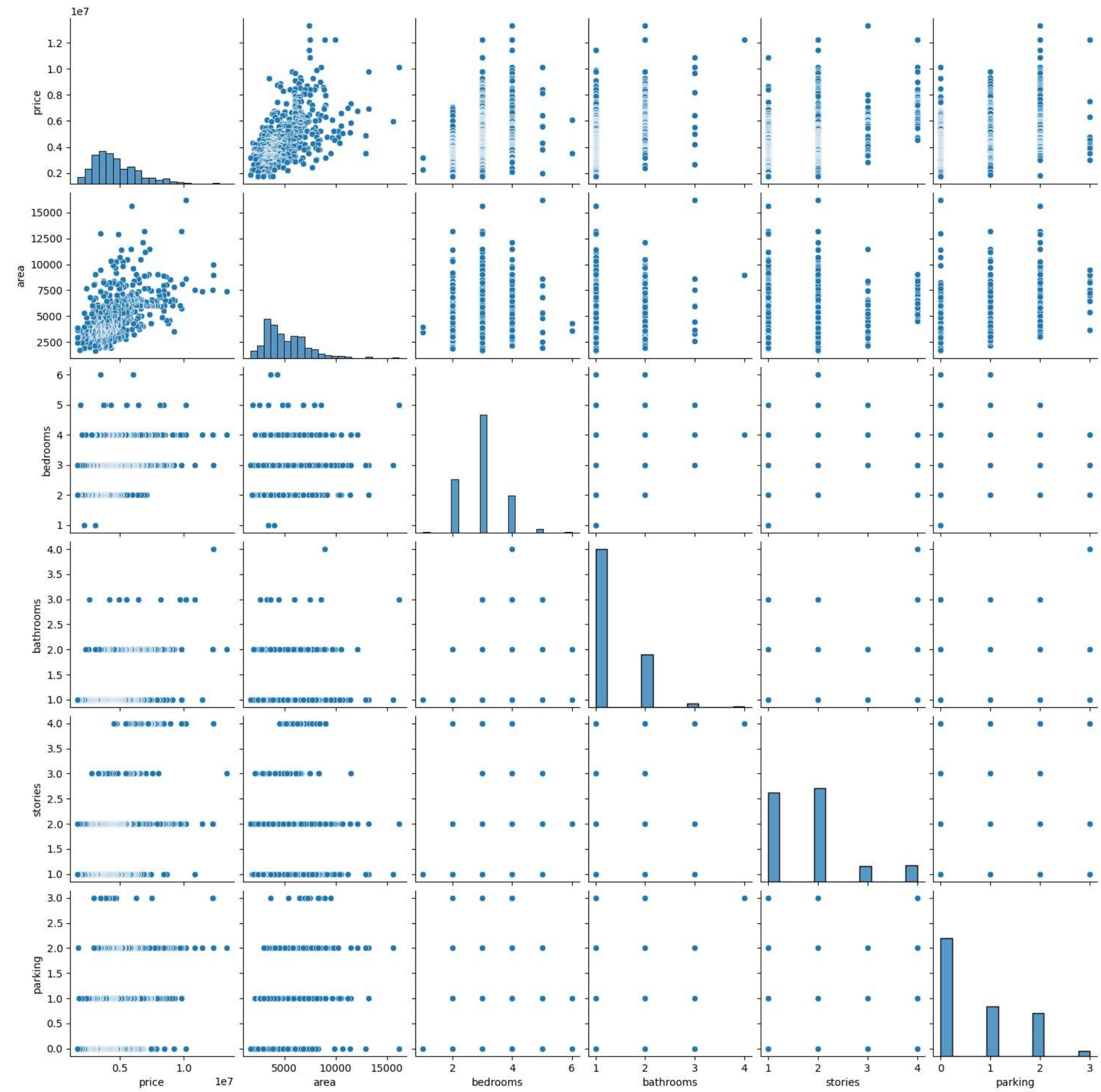
```python
print(df.isnull().sum())
```

```
price               0
area                0
bedrooms            0
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64
```

```
plt1 =
sns.boxplot(df['price'],
ax = axs[0,0]) plt2 =
sns.boxplot(df['area'],
ax = axs[0,1]) plt3 =
sns.boxplot(df['bedrooms
'], ax = axs[0,2]) plt1
=
sns.boxplot(df['bathroom
s'], ax = axs[1,0]) plt2
=
sns.boxplot(df['stories'
], ax = axs[1,1]) plt3 =
sns.boxplot(df['parking'
], ax = axs[1,2])
plt.tight_layout()
```




```
sns.pairplot(df)
plt.show()
```

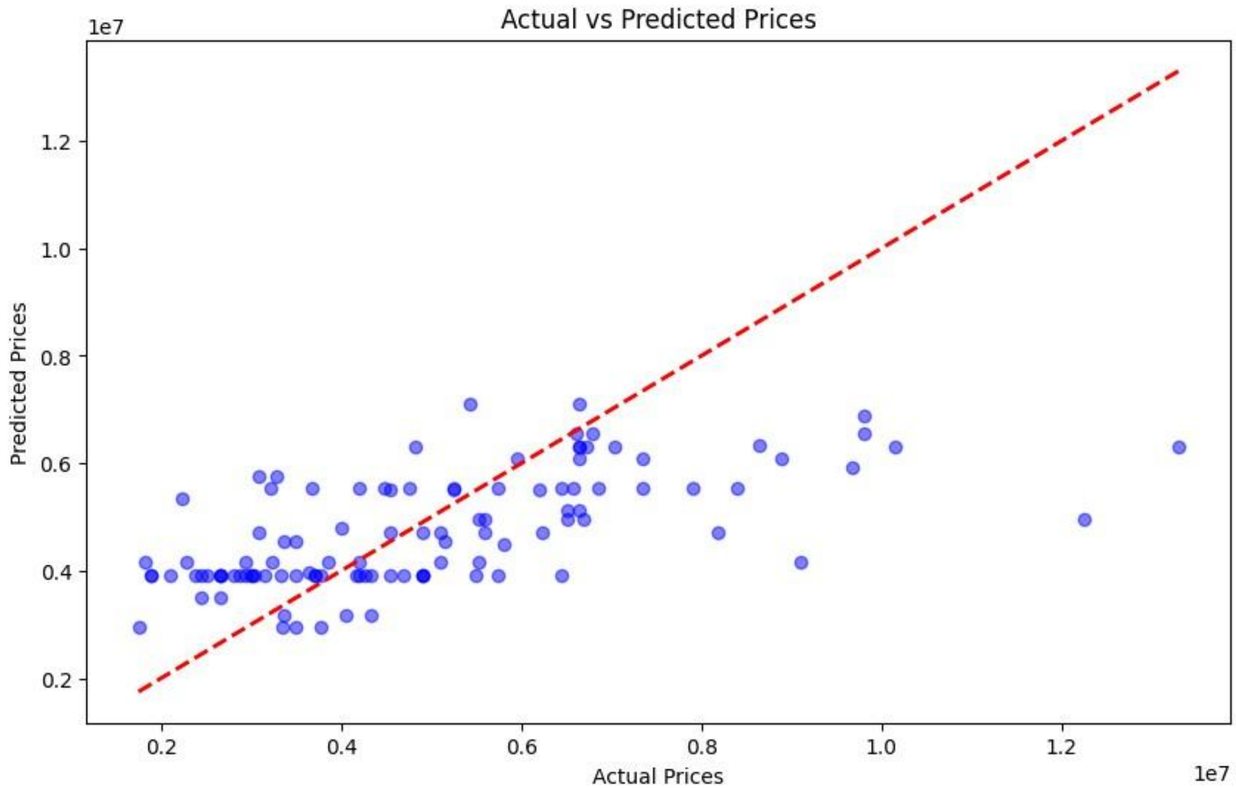



```
# Encoding Categorical Variables features = ['mainroad',
'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
'prefarea'] df_encoded = pd.get_dummies(df[features],
drop_first=True) # Adding Target Variable df_encoded['price'] =
df['price'] X = df_encoded.drop('price', axis=1)
y = df_encoded['price'] X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2, random_state=42) model =
LinearRegression() model.fit(X_train, y_train) y_pred =
model.predict(X_test)
```

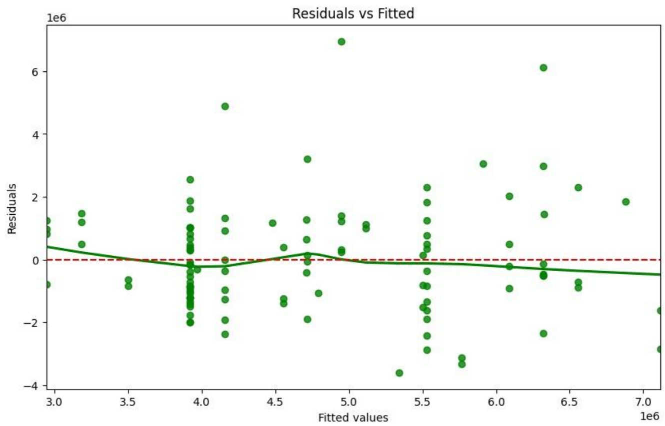
```python
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print(f'Mean Squared Error: {mse}')
print(f'R-squared: {r2}')
```

```
Mean Squared Error: 3254189950926.765
R-squared: 0.35618859916519907
```

```python
plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred, color='blue', alpha=0.5)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2)
plt.title('Actual vs Predicted Prices')
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.show()
```




```python
plt.figure(figsize=(10, 6)) sns.residplot(x=y_pred, y=y_test - y_pred, lowess=True, color='g') plt.title('Residuals vs Fitted') plt.xlabel('Fitted values') plt.ylabel('Residuals') plt.axhline(0, linestyle='--', color='red') plt.show()
```



Residuals vs Fitted
1e6
6
4
2
0
−2
−4
Residuals
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
1e6
Fitted values